

Notes: Heckman, Ichimura, Smith & Todd (Econometrica, 1998)

Characterizing Selection Bias Using Experimental Data

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1 Big picture

- **Question.** How can we characterize selection bias when nonexperimental comparison groups are used to evaluate social programs?
- **Design.** The paper uses a randomized JTPA experiment as a benchmark and compares experimental controls to eligible nonparticipants and SIPP comparison samples.
- **Contribution.** It estimates the bias function semiparametrically and tests identifying assumptions behind matching, selection models, and difference-in-differences.
- **Main findings.** Matching reduces but does not eliminate bias; classical inverse-Mills selection is misspecified; index-sufficient selection and conditional DID are more promising; data quality is central.

2 Data and measurement

2.1 Experimental benchmark and comparison groups

The experiment is the National JTPA Study. Random assignment occurred after individuals applied, were found eligible, and were accepted. Controls were denied JTPA services for 18 months. The nonexperimental ENP group consists of eligible nonparticipants in the same four labor markets, surveyed with the same instrument. SIPP samples are used to study what happens when geography and survey design differ.

Interpretation. The experimental controls identify the untreated counterfactual for accepted applicants, allowing the authors to measure nonexperimental bias directly.

2.2 Outcomes, covariates, and participation probability

The main outcome is quarterly earnings expressed in monthly dollars. Covariates include training center, race, age, schooling, marital status, children, local unemployment, recent labor-force transitions, job spells, prior training, household structure, and earnings. The propensity score $P(X) = Pr(D = 1 | X)$ is estimated from a weighted logit comparing controls to ENPs.

Interpretation. Recent labor-force histories, not only lagged earnings, are crucial predictors of program participation.

3 Models and empirical specifications

3.1 Selection bias

$$B(X) = E(Y_0 | X, D = 1) - E(Y_0 | X, D = 0).$$

Matching on X or $P(X)$ is valid only if this bias is zero over common support.

Interpretation. The paper estimates $B(X)$ instead of assuming it away.

3.2 Conventional bias decomposition

$$B = B_1 + B_2 + B_3,$$

where B_1 is support mismatch, B_2 is density-weighting mismatch, and B_3 is selection bias over common support.

Interpretation. A raw mean difference is not a clean measure of selection bias.

3.3 Semiparametric bias model

$$Y_{it} = X_{it}\beta + D_i K_{1t}(P_i) + (1 - D_i)K_{0t}(P_i) + \varepsilon_{it}.$$

The functions K_{1t} and K_{0t} are estimated by local linear regression; their difference gives $B_t(P)$.

Interpretation. The model estimates bias flexibly as a function of participation probability.

3.4 Conditional difference-in-differences

$$B_t(P) - B_{t'}(P) = 0.$$

If the bias function is time-invariant conditional on P , differencing pre- and post-periods removes bias.

Interpretation. The panel structure of the data allows the authors to test this assumption directly.

4 Identification and empirical strategy

- Random assignment provides the benchmark untreated potential outcome for accepted applicants.
- Comparing controls and ENPs isolates nonexperimental selection bias because both groups are untreated.
- Common support and common weighting separate genuine selection bias from sample-design artifacts.
- Matching is tested by conditional mean-independence tests.
- Index sufficiency is tested by asking whether additional covariates change $B(P)$.
- Conditional DID is tested by asking whether $B_t(P)$ is stable over time.
- SIPP comparisons diagnose the importance of geography, questionnaire design, and data quality.

5 Tables and figures

5.1 Figure 1: Prototypical selection model; Figure 2: Density of estimated participation probability

Read: Figure 1 shows the classical shape of bias in a normal selection model. Figure 2 shows overlap and mismatch in the estimated $P(X)$ distributions for controls and ENPs.

Economic message. Bias is a function over support, not a single number.

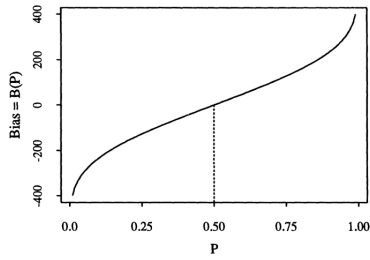
TABLE 1
DEFINITION OF VARIABLES

Variable Name	Description
Training Center: Corpus Christi, Fort Wayne, Jersey City, Providence.	Indicator variables for the geographic location of the individual.
Race and Ethnicity: black, white, Hispanic.	Indicator variables for the race/ethnicity of the individual. Individuals who reported Asian or "other" were included in the Hispanic category in <i>R</i> but not in <i>Z</i> .
Age: age 22–29, age 30–39, age 40–49, age 50–54.	Indicator variables for the age of the individual calculated using the average age in years of the individual within the quarter of the observation.
Education: less than 10th grade, 10–11th grade, 12th grade, 1–3 years of college, 4 or more years of college.	Indicator variables for the educational attainment of the individual at the time of random assignment or eligibility determination. Missing values are imputed. ^a
Marital Status: currently married, last married 1–12 months before RA/EL, last married > 12 months before RA/EL, single, never married at RA/EL.	Indicator variables for marital status at the time of random assignment or eligibility determination (RA/EL). Missing values are imputed. ^a
Children less than 6 years of age	Indicator variable for the presence of young children in the household at the time of the baseline interview. Missing values are imputed. ^a
Calendar Quarter: quarter 1, quarter 2, quarter 3, quarter 4.	Indicator variables for the calendar quarter for the observations. Quarter 1 refers to January, February, and March etc. If an observation overlaps two quarters, then the variable takes on fractional values.
Calendar Year: year 1987, year 1988, year 1989, year 1990.	Indicator variables for the calendar year of the observation. If the observation overlaps two years, then the year indicators take on fractional values.
Local Unemployment Rate (Sources: U.S. Department of Labor's publication "Labor Force, Employment, and Unemployment Estimates for States, Labor Market Areas, Counties, and Selected Cities" for the years 1986–1991 provide the unemployment rates. Population weights are obtained from annual total population data available in the U.S. Department of Commerce's Regional Economic Information System (REIS)).	This variable gives the monthly unemployment rate. The data are published at the county and metropolitan area levels. We calculate the unemployment rate as a population-weighted average of the unemployment rates of the counties and metropolitan areas served by each of the four training centers in the JTPA data.

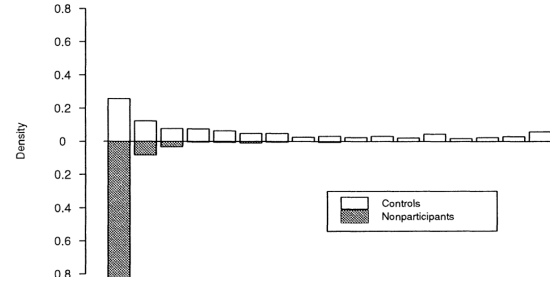
TABLE 1—Continued

Variable Name	Description
Labor Force Status Transition: employed → employed, unemployed → employed, OLF → employed, employed → unemployed, unemployed → unemployed, OLF → unemployed, employed → OLF, unemployed → OLF, OLF → OLF.	The two most recent labor force statuses during the period composed of the month of random assignment or eligibility determination and the six preceding months define a set of nine labor force status patterns. In each case, the second status is that in the month of random assignment or eligibility determination and the first status (if different) is the most recent preceding status. Repeated patterns such as "employed → employed" indicate persons in the same labor force status for all seven months. Missing values are imputed. ^a
Number of Persons in the Household	Continuous variable indicating the number of persons in the individual's household as of the baseline interview. Missing values are imputed. ^a
Earnings in the Month of Random Assignment or Eligibility Determination	Self-reported monthly earnings in the month of random assignment or eligibility determination from the baseline survey. Persons for whom the survey covers only a part of the month have their responses scaled up to a full month.
Ever had Vocational Training	Indicator variable for whether the respondent ever had vocational or technical training as of the baseline interview date, excluding courses taken while in high school. Missing values are imputed. ^a
Currently Receiving Vocational Training	Indicator variable for current receipt of vocational or technical training as of the baseline interview. Excludes courses taken in high school. Missing values are imputed. ^a
Number of Job Spells in the 18 Months Prior to Random Assignment or Eligibility Determination: zero, one, two, more than two.	Categories for the number of full or partial job (not employment) spells experienced during the 18 months prior to random assignment or eligibility determination. Missing values are imputed. ^a
Work Experience	Continuous variable indicating months of work experience prior to random assignment or eligibility determination. It is calculated using the Mincer method, (age-education-6)*12, for the period prior to our data, adding in actual experience in months for the five years prior to RA/EL.

^aAn appendix available upon request from the authors describes the imputation procedure for these variables.



(a) Figure 1: prototypical selection model

(b) Figure 2: density of estimated P

5.2 Table I: Definition of variables; Table II: Bias by labor-force transition cells

Read: Table I documents rich covariates. Table II shows that recent labor-force transitions strongly predict participation and that cell-level bias can be large.

Economic message. Measurement of recent labor-market histories is central to constructing a credible comparison group.

TABLE II
ESTIMATED BIAS BY LABOR FORCE STATUS TRANSITION CELLS AND THE PROBABILITY OF
PARTICIPATION AND ITS LOGIT BY LABOR FORCE STATUS, CRUDE RATES,
AND RATES IMPLIED BY LOGIT
Quarterly Earnings Expressed in Monthly Dollars, Experimental Control and
Eligible Nonparticipant (ENP) Samples, Adult Males, 508 Controls and 388 ENPs

Cell	Percentage of Controls in Cell	Percentage of ENPs in Cell	Estimated Bias in Cell ^a	Difference in Population Program Participation Rates ^b	Average Derivative from Logit ^b	Coefficient From Logit ^b	Difference in Logits of Program Participation Rates ^b
Employed → Employed	21.16	73.41	-421 (78)	*	*	*	*
Unemployed → Employed	10.79	4.16	-474 (228)	0.038	0.033	1.518 (0.416)	1.703
OLF → Employed	4.77	1.11	-128 (327)	0.061	0.012	0.787 (0.761)	2.139
Employed → Unemployed	27.39	6.65	-148 (188)	0.158	0.079	2.465 (0.456)	3.113
Unemployed → Unemployed	17.43	4.16	420 (69)	0.105	0.093	2.668 (0.583)	2.671
OLF → Unemployed	5.81	0.55	230 (207)	0.109	0.142	3.272 (0.597)	2.715
Employed → OLF	5.60	1.94	398 (209)	0.119	0.085	2.552 (0.608)	2.801
Unemployed → OLF	1.45	1.39	381 (92)	0.072	0.069	2.302 (0.875)	2.298
OLF → OLF	5.60	6.65	297 (157)	0.017	-0.002	-0.154 (0.609)	1.090

^aThis column gives the mean difference in monthly earnings of the experimental controls and eligible nonparticipants conditional on labor force status transition patterns in the six months prior to random assignment (see Table I for the definition of the labor force transition categories). The mean is calculated over the 18 months after the date of random assignment/eligibility determination.

^bThese columns give differences in the population participation rates and in the logits of the population participation rate, relative to the Employed → Employed cell.

a common support of $P(X)$. For the sample of controls, the histogram of $P(X)$ values has support over the entire $[0, 1]$ interval. Surprisingly, however, the mode of the distribution of $P(X)$ for controls is near zero. Many controls have a low estimated probability of participation. In the sample of ENPs, the support of $P(X)$ is concentrated in the interval $[0, 0.225]$. Thus, the bias measure $\bar{B}_{S_P}$, which is the bias defined conditional on $P(X)$ rather than X , is defined only over a fairly limited interval. As a result of this restriction on the support, any nonexperimental evaluation can nonparametrically estimate program impacts defined only over this interval. As we demonstrate below, the difference between the distributions of the estimated values of P has important implications for understanding the sources of selection bias as conventionally measured. Before presenting this decomposition, we first develop some econometric tools that are used to generate many of the empirical results reported in this paper.

Table II: bias by labor-force transition cells

5.3 Table III and Table V: Participation logit and bias decomposition

Read: Table III estimates $P(X)$. Table V decomposes conventional bias into support mismatch, weighting differences, and genuine selection bias.

Economic message. Much raw bias is generated by support and weighting problems, not just unobserved selection.

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SELECTION BIAS

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TABLE III
COEFFICIENT ESTIMATES AND p VALUES FROM WEIGHTED PARTICIPATION LOGIT^a
BEST PREDICTOR MODEL FOR THE PROBABILITY OF PARTICIPATION^b
Experimental Control and Eligible Nonparticipant (ENP) Samples
Dependent Variable: 1 for Experimental Control, 0 for Eligible Nonparticipant
Adult Males, 508 Controls and 388 ENPs

Variables	Coeff	Std Error	p Value ^c
Intercept	-5.07	0.83	0.0000
Fort Wayne, IN	2.45	0.41	0.0000
Jersey City, NJ	0.66	0.43	0.1273
Providence, RI	2.19	0.44	0.0000
Black	0.49	0.33	0.1333
Hispanic	0.43	0.40	0.2837
Other race/ethnicity	0.61	0.55	0.2653
Age 30 to 39	-0.50	0.30	0.0926
Age 40 to 49	-0.60	0.38	0.1115
Age 50 to 54	-0.29	0.62	0.6361
Fewer than 10 years schooling	-0.83	0.40	0.0397
10–11 years schooling	0.66	0.34	0.0510
13–15 years schooling	0.90	0.35	0.0096
16 or more years schooling	-1.38	0.54	0.0101
Last married 1–12 months prior to RA/EL ^d	0.42	0.80	0.5995
Last married > 12 months prior to RA/EL	-0.03	0.61	0.9648
Single, never married at RA/EL	0.71	0.36	0.0498
Child age less than 6 present in household	-0.16	0.38	0.6761
Unemployed → Employed	1.52	0.42	0.0003
OLF → Employed	0.79	0.76	0.3016
Employed → Unemployed	2.46	0.46	0.0000
Unemployed → Unemployed	2.67	0.58	0.0000
OLF → Unemployed	3.27	0.60	0.0000
Employed → OLF	2.55	0.61	0.0000
Unemployed → OLF	2.30	0.87	0.0085
OLF → OLF	-0.15	0.61	0.8002
One job in 18 months prior to RA/EL	0.41	0.39	0.2894
Two jobs in 18 months prior to RA/EL	0.57	0.50	0.2600
More than two jobs in 18 months prior to RA/EL	1.87	0.52	0.0003
Enrolled in vocational training at RA/EL	1.94	0.62	0.0019
Ever had vocational training?	-0.28	0.32	0.3815
Total number of household members	-0.25	0.10	0.0134
Earnings in the month of RA/EL	-0.00	0.00	0.0000

^aWeights are used in the estimation procedure to account for choice-based sampled data. It is assumed that in a random sample Controls represent 3% and ENPs 97% of the eligible population.

^bThe omitted training center is Corpus Christi, TX; the omitted race is white; the omitted age group is 22–29; the omitted schooling category is twelve years; the omitted marital status is currently married at RA/EL; the omitted labor force transition pattern is Employed → Employed; the omitted number of job spells in the 18 months prior to RA/EL is zero.

^cReported p -values are for two-tailed tests of the null hypotheses that the true coefficient equals zero.

^dRA/EL indicates the month of random assignment (RA) for the experimental controls and the month of eligibility (EL) for Eligible Nonparticipants (ENPs).

(a) Table III: participation logit

TABLE V
DECOMPOSITION OF MEAN SELECTION BIAS FOR THE BEST PREDICTOR MODEL FOR THE
PROBABILITY OF PROGRAM PARTICIPATION^a
Experimental Control and Elig. Nonparticipant (ENP) Samples, Adult Males,
508 Controls and 388 ENPs

Quarter	(1) Mean Difference ^b ($\hat{B}$)	(2) Nonoverlap Support ^c ($\hat{B}_1$)	(3) Density Weighting ($\hat{B}_2$)	(4) Selection Bias ($\hat{B}_3$)	(5) Average Bias ($\hat{B}_{S_P}$)	(6) Experimental Treatment Impact	(7) Average Bias ($\hat{B}_{S_P}$) as of % of Treatment Impact ^d
Qtr1	-420 (38)	190[-45%] (31)	-627[149%] (32)	17[-4%] (34)	29 (63)	5 (30)	566%
Qtr2	-352 (47)	209[-59%] (41)	-58[165%] (45)	19[-6%] (35)	32 (65)	37 (33)	88%
Qtr3	-343 (55)	221[-65%] (39)	-576[168%] (50)	12[-3%] (43)	20 (79)	57 (34)	35%
Qtr4	-294 (57)	234[-80%] (40)	-568[194%] (46)	41[-14%] (42)	68 (79)	60 (34)	114%
Qtr5	-311 (57)	232[-75%] (40)	-576[185%] (51)	33[-10%] (41)	54 (77)	44 (35)	121%
Qtr6	-334 (63)	223[-67%] (45)	-573[172%] (51)	16[-5%] (44)	27 (81)	61 (34)	44%
Average of 1 to 6	-342 (47)	218[-64%] (38)	-584[170%] (41)	23[-7%] (33)	38 (63)	44 (14)	87%

^aThe best predictor model for the probability of program participation includes training center indicators, race, age, education, marital status, children aged less than 6, labor force status transitions, job spells, current and past vocational training, total number of household members, and earnings in the month of random assignment or eligibility determination. (See Table III for model estimates and Appendix C for prediction rate comparisons.)

^bThe percentage of the mean difference attributable to each component appears in square brackets in the appropriate column. Bootstrapped standard errors based on 50 replications with 100% sampling appear in parentheses.

^cA 2% trimming rule was used in determining the overlapping support region, and a 0.06 fixed bandwidth was used for the nonparametric estimates. (See Appendix A for details.) Proportion of controls in the overlap region $S_P = 0.60$, proportion of ENPs in $S_P = 0.96$.

^dThe final column gives the ratio of the absolute value of $\hat{B}_{S_P}$ to the absolute value of the experimental impact estimate, times 100. The experimental impact estimate is based on the full treatment and control sample.

presents $\hat{B}_{S_P}$ ($\hat{B}_{S_P}$ evaluated with $X = P$), the selection bias for those in the overlap set S_P . Column (6) presents the experimental impact estimate calculated using the full control and treatment group samples while column (7) expresses $\hat{B}_{S_P}$ as a fraction of the experimental program impact estimate. All of the values in the table are reported as monthly dollars. Thus the first row and first column of Table V reports a mean earnings difference of -\$420 per month over the three months of the first quarter after random assignment. The percentages of controls and ENPs in the common support region for P_i are reported in the table notes.

A remarkable feature of the estimates in Table V is that for the overall 18 month earnings measure, terms $\hat{B}_1$ and $\hat{B}_2$ are substantially larger than the selection bias component $\hat{B}_3$. The selection bias is a small fraction (only 7%) of the conventional measure of selection bias and is not statistically significantly different from zero.⁴³ These results on the bias for the overall impact of the

⁴³ For adult women and for youth the estimated selection bias is proportionately higher, although the conventional measure $\hat{B}$ is lower than for adult males. For adult women and youth the bias measures $\hat{B}$ and $\hat{B}_3$ are of the same order of magnitude. These results are reported in Heckman,

(b) Table V: bias decomposition

5.4 Tables VIA–VIB and Figure 3: Conditional mean independence and pointwise bias

Read: Tables VIA and VIB test matching assumptions; Figure 3 shows nonparametric pointwise bias and the poor fit of the inverse-Mills model.

Economic message. Matching may reduce bias but does not justify assuming $B(P) = 0$.

TABLE VIA

TESTS OF CONDITIONAL MEAN INDEPENDENCE OF EARNINGS AND RESIDUALS
BASED ON ASYMPTOTIC STANDARD ERRORS WITHOUT ADJUSTMENT
FOR ESTIMATION OF β^a
Experimental Controls and Elig. Nonparticipant (ENP) Samples
Adult Males, 508 Controls and 388 ENPs

p -Values from Tests of Conditional Mean Independence of Earnings ^b $H_0: E(Y_0 P, D=1) = E(Y_0 P, D=0)$		
Value of P	Joint Test for Quarters $t = 1$ to $t = 6^c$	Joint Test for Quarters $t = -6$ to $t = -1$
0.0025	0.0000	0.0242
0.005	0.0002	0.0803
0.01	0.0042	0.2416
0.02	0.1224	0.1919
0.03	0.4363	0.1238
0.04	0.7659	0.1585
0.05	0.9423	0.3271
0.10	0.1678	0.8464
Joint	0.0000	0.0159

p -Values from Tests of Conditional Mean Independence of Residuals ^b $H_0: E(U_0 P, D=1) = E(U_0 P, D=0)$		
Value of P	Joint Test for Quarters $t = 1$ to $t = 6$	Joint Test for Quarters $t = -6$ to $t = -1$
0.0025	0.0002	0.0293
0.005	0.0004	0.0815
0.01	0.0040	0.2586
0.02	0.2056	0.4078
0.03	0.6563	0.5680
0.04	0.9060	0.7177
0.05	0.9885	0.8064
0.10	0.2591	0.7456
Joint	0.0001	0.2515

^aDensities were estimated using a biweight kernel and using the fixed bandwidth proposed in Silverman (1986) (defined in Appendix A, Section A.2). Conditional means were estimated by local linear regression using a fixed bandwidth of 0.06 and a biweight kernel. (See Appendix A, Section A.1 for a description of local linear regression and Section A.6 for a description of the test procedure.)

^bFinal row presents the p -value from a joint test.

^cThe number of observations within one bandwidth of $P = 0.0025$ in quarter 1 are 140 controls and 328 ENPs. For other P points, the numbers of observations are the following: 143 controls, 331 ENPs ($P = 0.005$), 150 controls and 336 ENPs ($P = 0.01$), 158 controls and 345 ENPs ($P = 0.02$), 170 controls and 350 ENPs ($P = 0.03$), 184 controls and 353 ENPs ($P = 0.04$), 198 controls and 355 ENPs ($P = 0.05$), and 120 controls and 52 ENPs ($P = 0.1$). The number of observations in other quarters are similar, but vary slightly because of the unbalanced panel data.

(a) Table VIA

TABLE VIB

TESTS OF CONDITIONAL MEAN INDEPENDENCE OF RESIDUALS BASED ON
ASYMPTOTIC STANDARD ERRORS WITH ADJUSTMENT FOR
ESTIMATION OF β^a
Experimental Controls and Elig. Nonparticipant (ENP) Samples
Adult Males, 508 Controls and 388 ENPs

p -Values from Tests of Conditional Mean Independence of Residuals ^b $H_0: E(U_0 P, D=1) = E(U_0 P, D=0)$		
Value of P	Joint Test for Quarters $t = 1$ to $t = 6$	Joint Test for Quarters $t = -6$ to $t = -1$
0.0025	0.0955	0.6421
0.005	0.1129	0.7346
0.01	0.2170	0.7773
0.02	0.5925	0.6738
0.03	0.8289	0.7808
0.04	0.9563	0.8726
0.05	0.9939	0.9042
0.10	0.5790	0.9253
Joint	1.0000	1.0000

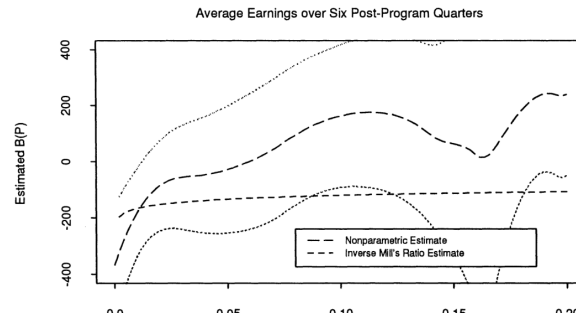
^aDensities were estimated using a biweight kernel and using the fixed bandwidth proposed in Silverman (1986) (defined in Appendix A, Section A.2). Conditional means were estimated by local linear regression using a fixed bandwidth of 0.06 and a biweight kernel. (See Appendix A, Section A.1 for a description of local linear regression and Section A.6 for a description of the test procedure.)

^bFinal row presents the p -value from a joint test.

tion P located at least one bandwidth apart, so that the test statistics are statistically independent. The top portion of Table VIA reports tests of hypothesis (7). The relevant period over which the test should be performed is the post-random assignment period ($t = 1, \dots, 6$) since it is post-entry time periods on which the program would be evaluated. For the sake of completeness, however, we also record the test results for the pre-random assignment period ($t = -1, \dots, -6$).⁴⁴ The bottom portion of the table reports tests of hypothesis (21). Hypothesis (7), which justifies matching on P , is decisively rejected. In addition, hypothesis (21) is rejected, so regression-adjusted matching is also inconsistent with our data. When second order-adjusted standard errors are used that account for the estimation of β , as in Table VIB, the evidence is less clear cut. However, the pointwise bias is large (see Figure 3 for bias from the best-predictor P) and it seems inappropriate to ignore this bias and accept the null of no selection bias when an asymptotically-equivalent test of the same hypothesis rejects it. Table VII reports the pointwise bias estimates at deciles of the distribution of P for controls. The bias is large, negative, and statistically significant at low values of P and large and positive at high values of P , which is inconsistent with the null hypothesis that matching is a valid estimator.

⁴⁴ The same inferences are found when we test over all 12 periods although such a test is not especially interesting for judging the performance of matching as an evaluation estimator on post-random-assignment data.

(b) Table VIB

Figure 3: pointwise bias $B(P)$

5.5 Tables VII–VIII: Decile bias and index sufficiency

Read: Table VII reports bias by decile of P ; Table VIII tests whether covariates beyond P change the bias function.

Economic message. Index sufficiency is plausible but not proven with high power.

TABLE VIII
 p -VALUES FROM TESTS OF INDEX SUFFICIENCY^a
 Experimental Controls and Elig. Nonparticipant (ENP) Samples
 Best Predictor Model for the Probability of Program Participation
 Adult Males, 508 Controls and 388 ENPs

Tests by Race and Ethnicity ^b		
Value of P	Joint Test for Quarters $t = 1$ to $t = 6$	Joint Test for Quarters $t = -6$ to $t = -1$
0.002	0.4229	0.4376
0.039	0.5213	0.7867
0.076	0.2268	0.5307
0.113	0.6526	0.2827
0.150	0.0175	0.2440
Joint ^c	0.0421	0.3983
Tests by Training Center ^b		
Value of P	Joint Test for Quarters $t = 1$ to $t = 6$	Joint Test for Quarters $t = -6$ to $t = -1$
0.008	0.4942	0.8503
0.026	0.3404	0.1392
0.044	0.0952	0.0230
0.062	0.0925	0.0626
0.080	0.4062	0.2633
Joint ^c	0.4667	0.5959
Tests by Years of Schooling Categories ^b		
Value of P	Joint Test for Quarters $t = 1$ to $t = 6$	Joint Test for Quarters $t = -6$ to $t = -1$
0.003	0.4717	0.4646
0.022	0.5736	0.2842
0.042	0.2576	0.0967
0.061	0.0792	0.0188
0.080	0.0967	0.0964
Joint ^c	0.1718	0.1686

^aDensities were estimated using a biweight kernel and using the fixed bandwidth proposed in Silverman (1986) (defined in Appendix A, Section A.2). Conditional means were estimated by local linear regression using a fixed bandwidth of 0.06 and a biweight kernel. (See Appendix A, Section A.1 for a description of local linear regression and Section A.6 for a description of the test procedure.) Standard errors used in the test are asymptotic and are not adjusted for higher order terms (as described in Appendix A, Section A.6). When adjustment is made for estimation of β , the estimated standard errors are substantially larger.

^bThe tests by race and ethnicity include "White" and "Black" groups. The tests by training center include "Fort Wayne," "Jersey City," and "Providence." The tests by years of schooling category include "Fewer than 10 years of schooling," "10-11 years of schooling," "12 years of schooling," and "More than 12 years of schooling."

^cJoint tests shown include only a subset of the P points that are at least one bandwidth apart.

(a) Table VII

TABLE VIII
 p -VALUES FROM TESTS OF INDEX SUFFICIENCY^a
 Experimental Controls and Elig. Nonparticipant (ENP) Samples
 Best Predictor Model for the Probability of Program Participation
 Adult Males, 508 Controls and 388 ENPs

Tests by Race and Ethnicity ^b		
Value of P	Joint Test for Quarters $t = 1$ to $t = 6$	Joint Test for Quarters $t = -6$ to $t = -1$
0.002	0.4229	0.4376
0.039	0.5213	0.7867
0.076	0.2268	0.5307
0.113	0.6526	0.2827
0.150	0.0175	0.2440
Joint ^c	0.0421	0.3983
Tests by Training Center ^b		
Value of P	Joint Test for Quarters $t = 1$ to $t = 6$	Joint Test for Quarters $t = -6$ to $t = -1$
0.008	0.4942	0.8503
0.026	0.3404	0.1392
0.044	0.0952	0.0230
0.062	0.0925	0.0626
0.080	0.4062	0.2633
Joint ^c	0.4667	0.5959
Tests by Years of Schooling Categories ^b		
Value of P	Joint Test for Quarters $t = 1$ to $t = 6$	Joint Test for Quarters $t = -6$ to $t = -1$
0.003	0.4717	0.4646
0.022	0.5736	0.2842
0.042	0.2576	0.0967
0.061	0.0792	0.0188
0.080	0.0967	0.0964
Joint ^c	0.1718	0.1686

^aDensities were estimated using a biweight kernel and using the fixed bandwidth proposed in Silverman (1986) (defined in Appendix A, Section A.2). Conditional means were estimated by local linear regression using a fixed bandwidth of 0.06 and a biweight kernel. (See Appendix A, Section A.1 for a description of local linear regression and Section A.6 for a description of the test procedure.) Standard errors used in the test are asymptotic and are not adjusted for higher order terms (as described in Appendix A, Section A.6). When adjustment is made for estimation of β , the estimated standard errors are substantially larger.

^bThe tests by race and ethnicity include "White" and "Black" groups. The tests by training center include "Fort Wayne," "Jersey City," and "Providence." The tests by years of schooling category include "Fewer than 10 years of schooling," "10-11 years of schooling," "12 years of schooling," and "More than 12 years of schooling."

^cJoint tests shown include only a subset of the P points that are at least one bandwidth apart.

(b) Table VIII

5.6 Figure 4 and Table IX: Time path of $B_t(P)$ and DID tests

Read: Figure 4 plots bias over time; Table IX tests fixed-effect and DID restrictions.

Economic message. Conditional DID works best when bias is stable over time after conditioning on P .

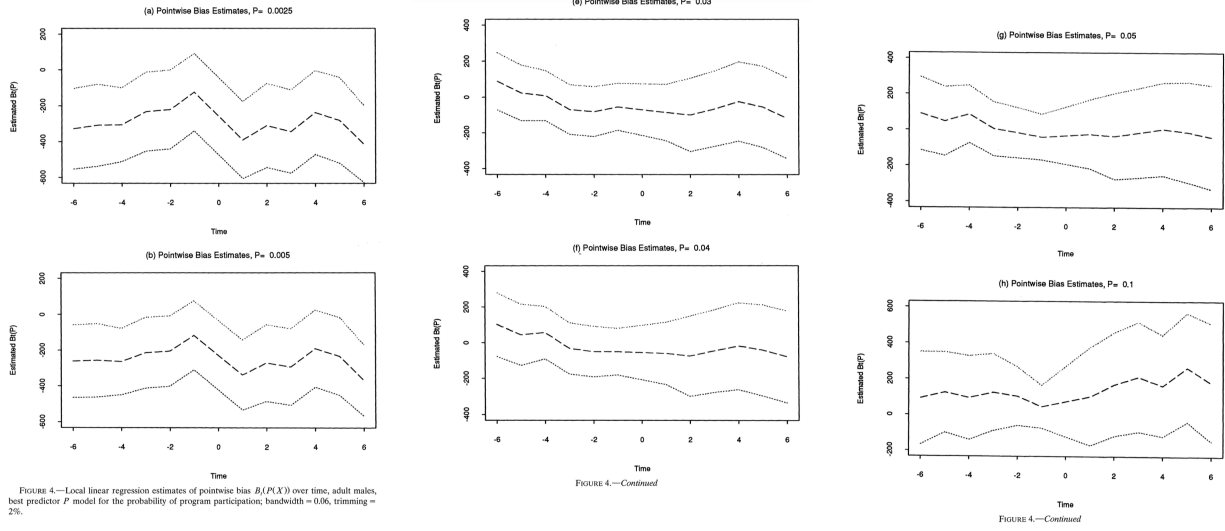


Figure 4: pointwise bias over time

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TABLE IX
p-VALUES FROM TESTS FOR FIXED EFFECT AND DIFFERENCE-IN-DIFFERENCES
SPECIFICATIONS FOR THE BIAS FUNCTION
All Tests are Symmetric Around Time $t = 0^{a,b}$
Adult Males, 508 Controls and 388 ENPs

Value of P (1)	Null tested jointly over $t \in \{1, 2, 3, 4, 5, 6\}^c$		
	Fixed Effect Test for Controls (2)	Fixed Effect Test for ENPs (3)	Difference-in- Differences Test (4)
0.0025	0.0922	0.8042	0.1221
0.0050	0.0967	0.8578	0.1291
0.0100	0.1148	0.8609	0.1688
0.0200	0.2158	0.3093	0.4279
0.0300	0.2327	0.0948	0.7579
0.0400	0.0807	0.0454	0.9353
0.0500	0.0047	0.0466	0.9707
0.1000	0.0057	0.1785	0.9914
Overall	0.1303	0.0019	0.8087
	Null tested jointly over $t \in \{1, 2, 3\}^c$		
	Fixed Effect Test for Controls (2)	Fixed Effect Test for ENPs (3)	Difference-in- Differences Test (4)
0.0025	0.3160	0.9417	0.4083
0.0050	0.3251	0.9832	0.4907
0.0100	0.3614	0.9949	0.7176
0.0200	0.4392	0.6269	0.9899
0.0300	0.3158	0.1877	0.9999
0.0400	0.1159	0.0785	1.0000
0.0500	0.0121	0.0680	0.9999
0.1000	0.0068	0.4363	0.9999
Overall	0.0255	0.0001	0.7008
	Null tested jointly over $t \in \{4, 5, 6\}^c$		
	Fixed Effect Test for Controls (2)	Fixed Effect Test for ENPs (3)	Difference-in- Differences Test (4)
0.0025	0.9456	0.9968	0.9386
0.0050	0.8779	0.9832	0.8424
0.0100	0.7217	0.9115	0.6266
0.0200	0.6256	0.8132	0.6123
0.0300	0.7353	0.8901	0.8768
0.0400	0.8467	0.9206	0.9772
0.0500	0.8522	0.9263	0.9906
0.1000	0.4140	0.9931	0.9966
Overall	0.9498	0.3141	0.9376

^aDensities were estimated using a bweight kernel and using the fixed bandwidth proposed in Silverman (1986) (defined in Appendix A, Section A.2). Conditional means were estimated by local linear regression using a fixed bandwidth of 0.06 and a bweight kernel. (See Appendix A, Section A.1 for a description of local linear regression and Section A.6 for a description of the test procedure.) Standard errors used in the test are asymptotic and are not adjusted for higher order terms (as described in Appendix A, Section A.6). When adjustment is made for estimation of β , the estimated standard errors are substantially larger.

^bNull hypothesis for fixed effects test for controls is $H_0: K_0(P) - K_{1,-}(P) = 0$; null hypothesis for fixed effect test for ENPs is $H_0: K_0(P) - K_{1,-}(P) = 0$; null hypothesis for difference-in-differences test is $H_0: [K_0(P) - K_{1,-}(P)] - [K_0(P) - K_{1,-}(P)] = 0$; where $(-)$ is a pre-program period t periods before random assignment or eligibility determination.

^cValues of P in the overall test are at least one bandwidth apart.

Table IX: tests for fixed-effect and DID restrictions

5.7 Tables X–XII: Bias by decile and alternative estimators

Read: Tables X–XII compare estimator behavior and sensitivity to outcome models.

Economic message. A small average bias can arise from offsetting components, so decomposition is essential.

TABLE X
DIFFERENCE-IN-DIFFERENCES AT DECILES OF THE CONTROL P DISTRIBUTION FOR THE BEST PREDICTOR MODEL FOR P^a
DIFFERENCES ARE SYMMETRIC AROUND TIME $t = 0^b$ QUARTERLY EARNINGS STATED IN MONTHLY DOLLARS
EXPERIMENTAL CONTROL AND ELIG. NONPARTICIPANT (ENP) SAMPLES, ADULT MALES, 508 CONTROLS AND 388 ENPs

Quarter	Decile of the Control Empirical Distribution of P^a (Decile boundaries shown in brackets)								
	1 (0.0002, 0.0023)	2 (0.0023, 0.0087)	3 (0.0087, 0.0152)	4 (0.0152, 0.0269)	5 (0.0269, 0.0410)	6 (0.0410, 0.0822)	7 (0.0822, 0.0983)	8 (0.0983, 0.1337)	9 (0.1337, 0.2534)
Qtr1	–269 (160)	–148 (125)	–63 (105)	–14 (91)	25 (92)	68 (106)	38 (140)	–112 (215)	858 (614)
Qtr2	–100 (164)	–42 (127)	–19 (108)	–23 (100)	–9 (114)	39 (147)	96 (161)	200 (177)	724 (720)
Qtr3	–119 (156)	–38 (131)	2 (115)	–2 (114)	2 (125)	65 (147)	126 (183)	413 (284)	627 (802)
Qtr4	54 (165)	63 (133)	26 (115)	–45 (113)	–34 (128)	99 (160)	113 (194)	141 (361)	–102 (681)
Qtr5	3 (186)	–2 (139)	–40 (114)	–46 (117)	–11 (157)	216 (209)	143 (196)	–350 (267)	38 (432)
Qtr6	–92 (172)	–139 (129)	–181 (114)	–149 (138)	–61 (182)	177 (228)	126 (217)	–168 (284)	–44 (401)
Average of 1 to 6	–87 (147)	–51 (114)	–46 (96)	–50 (94)	–14 (113)	111 (140)	107 (135)	20 (165)	350 (408)

^aThe best predictor model was used to estimate the probability of participation. It is given in Table III.

^bFor the nonparametric estimates, a fixed bandwidth of 0.06 and a bweight kernel function were used. (See Appendix A, Sections A.1 and A.4.2 for additional details concerning the estimation procedure.) Bootstrap standard errors are shown in parentheses. They are based on 50 replications with 100% sampling.

The deciles are based on the distribution of control probabilities of participation in the region of overlapping support. (See footnote to Table VII.) There are too few eligible nonparticipating observations to estimate the bias reliably in the 10th decile.

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TABLE XII
SELECTION BIAS UNDER DIFFERENT OUTCOME EQUATION MODELS:
ESTIMATED BIAS FROM REGRESSION-ADJUSTED LOCAL LINEAR MATCHING ESTIMATOR^b
QUARTERLY EARNINGS STATED IN MONTHLY DOLLARS
EXPERIMENTAL CONTROL ($D = 1$) AND ELIG. NONPARTICIPANTS (ENP) ($D = 0$) SAMPLES
ADULT MALES, 508 CONTROLS AND 388 ENPs

Quarter	Baseline ^c	Model I ^c	Model II ^c	Model III ^c	Model IV ^c	Model V ^c	Method of Barrow, Cain and Goldberger ^d
Qtr1	39 (60)	33 (59)	32 (68)	38 (68)	33 (65)	63 (62)	150 (34)
Qtr2	39 (64)	37 (61)	33 (64)	38 (68)	32 (63)	63 (65)	126 (28)
Qtr3	21 (80)	29 (78)	5 (76)	14 (78)	4 (76)	40 (83)	82 (16)
Qtr4	65 (82)	80 (77)	40 (78)	54 (78)	41 (79)	82 (82)	125 (27)
Qtr5	50 (83)	64 (77)	32 (75)	44 (78)	28 (77)	66 (80)	142 (45)
Qtr6	17 (90)	37 (82)	–5 (81)	9 (85)	–7 (84)	38 (87)	108 (53)
Average of 1 to 6	39 (64)	47 (60)	23 (61)	33 (63)	22 (62)	59 (65)	134 (51)
As a % of impact	88%	107%	52%	76%	50%	135%	304%

^aThe regression-adjusted average bias is defined in Appendix A, Section A.5.3. In the estimation of the model, densities were estimated using a bweight kernel and the fixed bandwidth proposed in Silverman (1986) (defined in Appendix A, Section A.2). The bias function was estimated by local linear regression using a fixed bandwidth of 0.06 and a bweight kernel. The overlapping support region was determined using a 2% trimming rule and a bweight kernel function. (See Appendix A, Section A.1 for a description of local linear regression, and Section A.4.1 for the method used to determine the overlapping support region.)

^bBootstrap standard errors are shown in parentheses. They are based on 50 replications with 100% sampling.

^cBaseline outcome model includes dummy variables for specific training center, race or ethnicity, schooling, age, and previous training. It also includes work experience, local unemployment rate, an indicator for marital status, an indicator for presence of a child age less than six, and quarter and year indicators.

Outcome Model I includes no R variables and is equivalent to local linear P score matching.

Outcome Model II augments I with training center indicators, race or ethnicity, age, and quarter and year indicators.

Outcome Model III augments II with previous training, work experience, local unemployment rate, and presence of a child age less than six.

Outcome Model IV augments II with local unemployment rate, presence of a child age less than six, schooling, and a marital status indicator.

Outcome Model V augments the baseline model with labor force transition indicators.

^dThe Barrow, Cain, and Goldberger (1980) method is based on equation (15) in the text with the $X = (R, Z)$ the same as in the baseline model, and $B(X) = B(Z)$ and $E(U_0|X, D = 0) = E(U_0|R, D = 0) = E(U_0|Z)$ assumed linear. We impose a common support restriction in defining the sample used for estimation where observations are used with $P(Z) \in S_p$ for the baseline model. The bias is computed using the distribution of $P|D = 1$.

TABLE XI
COMPARISON OF ESTIMATED SELECTION BIAS UNDER ALTERNATIVE ESTIMATORS
OF PROGRAM IMPACTS FOR THE BEST PREDICTOR MODEL FOR $P^{a,b}$
QUARTERLY EARNINGS STATED IN MONTHLY DOLLARS
EXPERIMENTAL CONTROL ($D = 1$) AND ELIG. NONPARTICIPANT (ENP) ($D = 0$) SAMPLES
ADULT MALES, 508 CONTROLS AND 388 ENPs

Quarter	(1) Difference in Means	(2) Local Linear P Score Matching	(3) Regression- Adjusted Local Linear P Score Matching	(4) Difference-in- Differences Local Linear P Score Matching	(5) Difference-in- Differences Regression-Adjusted Local Linear P Score Matching
Qtr1	–418 (38)	33 (59)	39 (60)	97 (62)	104 (63)
Qtr2	–349 (47)	37 (61)	39 (64)	77 (89)	77 (92)
Qtr3	–337 (55)	29 (78)	21 (80)	90 (114)	74 (114)
Qtr4	–286 (57)	80 (77)	65 (82)	112 (90)	98 (91)
Qtr5	–305 (57)	64 (77)	50 (83)	19 (95)	–5 (99)
Qtr6	–328 (63)	37 (82)	17 (90)	4 (105)	–35 (111)
Average of 1 to 6	–337 (47)	47 (60)	39 (64)	67 (71)	52 (74)
As a % of impact	775%	107%	88%	153%	120%

^aThe best predictor model was used for the probability of participation. It is given in Table III.

^bFor the nonparametric estimates, a fixed bandwidth of 0.06 and a bweight kernel function were used. (See Appendix A, Sections A.1, A.4, and A.5 for additional details concerning the estimation procedure.) Bootstrap standard errors are shown in parentheses. They are based on 50 replications with 100% sampling.

(a) Table XI

(b) Table XII

5.8 Tables XV–XVII: Geography, treatment-effect dependence, and restricted support

Read: Table XV highlights geographic mismatch; Figure 5 and Table XVI examine P -dependent treatment impacts; Table XVII compares full and restricted support.

Economic message. Data design and target population determine what impact parameter is learned.

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TABLE XV
EFFECT OF GEOGRAPHY ON ESTIMATED BIAS
COMPARING CONTROLS AT TWO SITES TO ELIGIBLE NON-PARTICIPANTS AT TWO SITES
Earnings in the 18 Months After Random Assignment
Quarterly Earnings Expressed in Monthly Dollars
Elig. Nonparticipant (ENP) Sample at Corpus Christi and Fort Wayne
Experimental Control Sample at Jersey City and Providence
Adult Males, 149 Controls and 276 ENPs

Quarter	Difference in Means B	Local Linear Matching ^b $\hat{B}_{LL}$	Regression-Adjusted Local Linear Matching $\hat{B}_{LL}(adj)$	Difference-in- differences for Local Linear Matching	Difference-in- differences for Regression-Adjusted Local Linear Matching
Qtr1	–534 (53)	–203 (85)	–184 (110)	–143 (111)	–135 (126)
Qtr2	–504 (73)	–166 (107)	–154 (120)	–125 (118)	–72 (130)
Qtr3	–515 (78)	–177 (120)	–147 (127)	–73 (131)	–9 (141)
Qtr4	–485 (78)	–200 (121)	–164 (132)	–87 (141)	19 (151)
Qtr5	–527 (72)	–272 (127)	–211 (132)	–254 (160)	–136 (167)
Qtr6	–524 (75)	–281 (110)	–189 (112)	–257 (162)	–82 (165)
Average of 1 to 6	–515 (63)	–216 (95)	–175 (108)	–157 (110)	–69 (123)
As a % of impact	1183%	497%	402%	360%	159%

^a 2% trimming is used to estimate the overlapping support region. A fixed bandwidth of 0.06 is used for the nonparametric estimates. (See Appendix A for more details on the estimation procedure.) Bootstrap standard errors are shown in parentheses. They are based on 50 replications with 100% sampling.

bias, and to evaluate the effectiveness of econometric methods in reducing the bias, we scramble the ENP-control data and mismatch by geography within these samples. Since all observations are administered the same questionnaire, this enables us to estimate a pure geographic mismatch effect. Table XV reports the result of matching ENPs ($D = 0$) in two training centers to controls ($D = 1$) from two other training centers. For three of the estimators, the bias B in Table XV is two or three times as large as the bias in the geographically-aligned data (compare with the results in Table XI). Matching and regression-adjusted matching reduce, but by no means eliminate, the bias (compare the second and third columns of Table XV with the second and third columns of Table XI). When data are geographically misaligned, the difference-in-differences estimators generally perform better than the cross-sectional estimators. Geographic mismatch is an important source of bias in evaluating training programs.^{55,56}

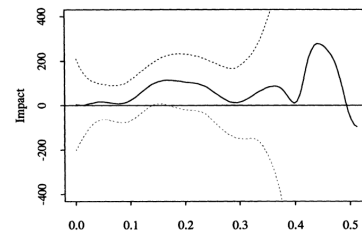
⁵⁵ Roselius (1996) builds on our analysis and creates a variety of SIPP samples using alternative definitions of region and city size. She finds substantial bias in all of her SIPP samples that is far in excess of the ENP-control bias reported in the text. Adjusting for labor market variables like the unemployment rate in the state or metropolitan statistical area does not reduce the bias she estimates.

⁵⁶ Smith (1995) uses other data sources and considers the consequences of alternative definitions of variables and survey instruments on the estimated bias.

(a) Table XV

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(b) Figure 5

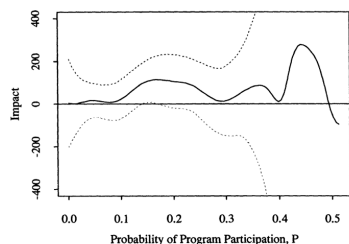


FIGURE 5.—Adult males, experimental treatments and controls, P dependence of treatment impacts, $E(Y_1 - Y_0 | P)$, bandwidth = 0.03.

TABLE XVI
 p -VALUES AND POINT ESTIMATES FROM TESTS OF P -DEPENDENCE OF
TREATMENT IMPACTS
 $H_0: E(Y_1 - Y_0 | P, D = 1) - E(Y_1 - Y_0 | D = 1) = 0$
Experimental Control and Treatment Samples
Average Monthly Earnings Over the First Six Quarters After Random
Assignment, Adult Males, 649 Controls and 1478 Treatments

Test Values of P	p -values ^a	Point Estimates ^b
0.00040	0.9607	-29
0.00081	0.9587	-31
0.00020	0.9529	-36
0.0024	0.9511	-38
0.0095	0.8952	-50
0.0159	0.7765	-47
0.0315	0.5760	-28
0.0494	0.7137	-25
0.0691	0.3765	-41
0.0970	0.6485	-38
0.1272	0.4745	38
0.1632	0.2647	74
0.2119	0.4271	57
0.2712	0.8116	-20

^aA bandwidth equal to 0.06 and a biweight kernel were used for the nonparametric estimates (see Appendix A, Sections A.1 for additional details concerning the estimation procedure). The distribution of the test statistic is chi-squared with one degree of freedom under the null.

^bValues shown are the difference between the conditional and unconditional estimated means.

(a) Table XVI

TABLE XVII
COMPARISON OF MONTHLY IMPACTS ESTIMATED
OVER THE ENTIRE SUPPORT AND OVER THE RESTRICTED SUPPORT^a
Quarterly Earnings Expressed in Monthly Dollars
Experimental Control and Treatment Samples
Adult Males, 649 Controls and 1478 Treatments

Quarter	Estimated Impact Using Entire Support ^b	Estimated Impact Using Restricted Support	Difference
Qtr1	4 (30)	18 (33)	-15 (12)
Qtr2	26 (38)	42 (36)	-16 (11)
Qtr3	51 (36)	69 (36)	-18 (13)
Qtr4	57 (42)	85 (51)	-28 (15)
Qtr5	39 (39)	58 (36)	-18 (14)
Qtr6	49 (44)	71 (35)	-23 (15)
Average of 1 to 6	38 (16)	57 (16)	-20 (5)

^aBootstrap standard errors are shown in parentheses. They are based on 50 replications with 100% sampling.

^bIn our data the experimental control group was administered a long baseline survey that gathered five years of retrospective data while the experimental treatment group was not (see Appendix B). Since information on recent labor force status and on recent earnings is missing for treatments, we are only able to obtain coarse estimates of P for the treated group. In particular, we use the coarse II model described in the notes to Table XIII. The support region in the nonexperimental analysis is determined using the best predictor P model, so it is necessary to estimate which treatment group members would be excluded by the common support restriction in order to obtain impact estimates within the support region that would be estimated by a nonexperimental method. The adjusted treatment impacts were obtained as follows. For controls and treatments, we first divide the coarse P distribution into 20 equal-size bins, then within-bin treatment impacts are estimated. The average unadjusted impact estimate is obtained as the weighted average of the within-bin estimates, with weights given by the proportion of controls within each bin. The adjusted impact estimate is equal to the weighted average of the within-bin estimates, with the weights given by the proportion of controls within each bin after deleting controls whose values of P lie outside the overlap region.

Consider equation system (8) and suppose that index sufficiency characterizes the bias term and that

$$(23a) \quad E(Y_1 | Z, R, D = 1) = g_1(R) + E(U_1 | P(Z), D = 1)$$

and

$$(23b) \quad E(Y_0 | Z, R, D = 0) = g_0(R) + E(U_0 | P(Z), D = 0).$$

If there is at least one element in Z not in R that satisfies the conditions

$$(24a) \quad \lim_{Z \rightarrow Z^{c_1}} E(U_1 | P(Z), D = 1) = 0$$

and

$$(24b) \quad \lim_{Z \rightarrow Z^{c_0}} E(U_0 | P(Z), D = 0) = 0,$$

where Z^{c_0} and Z^{c_1} may be values or sets of values, and need not be the same sets of values, we can identify $g_1(R)$ and $g_0(R)$ following the argument in

(b) Table XVII

5.9 Tables XVIII–XIX: Bias components under alternative estimators

Read: These tables show how support restrictions, density weighting, and regression adjustment change estimated bias across estimators.

Economic message. The practical message is that support and weighting are as important as functional form.

bias so that it excludes bias arising from gaps in the common support and from differences in the weights applied to participant and comparison group samples over the region of common support.

Using a common support and a common set of weights applied to participant and comparison group samples goes a long way toward improving the performance of any econometric evaluation estimator. Table XVIII clearly demonstrates this point. Column (1) presents the raw bias ($\hat{B}$) quarter-by-quarter and overall using the means for the control and ENP samples. Column (2) shows how the bias is reduced simply by matching to the nearest neighbor using P . (Recall that nearest neighbors can be far apart.) Column (3) shows how the imposition of the common support condition improves the nearest-neighbor matching estimator. Quarter-by-quarter, there is a substantial reduction in bias. However, the overall average is slightly higher in (3). Column (4) presents estimates of the bias that arise from local linear matching (on P) while column (5) presents the estimates that arise from regression-adjusted local linear matching. Both procedures impose common support and common weighting and both improve over the raw mean or crude nearest-neighbor estimators.

TABLE XVIII
COMPARISON OF ESTIMATED MEAN BIAS
UNDER ALTERNATIVE ESTIMATORS OF MEAN PROGRAM IMPACTS^a
Quarterly Earnings Expressed in Monthly Dollars
Adult Males, 508 Experimental Controls and 388 Elig. Nonparticipants (ENPs)

Quarter	Difference in Means w/o Common Support (1) ^b	Nearest Neighbor w/o Common Support (2)	Nearest Neighbor w/ Common Support (3)	Local Linear Matching (4)	Regression-Adjusted Local Linear Matching (5)
Qtr1	-418 (38)	221 (56)	123 (67)	33 (59)	39 (60)
Qtr2	-349 (47)	-166 (151)	77 (83)	37 (61)	39 (64)
Qtr3	-337 (55)	-58 (206)	53 (96)	29 (78)	21 (80)
Qtr4	-286 (57)	161 (178)	86 (96)	80 (77)	65 (82)
Qtr5	-305 (57)	167 (196)	87 (100)	64 (77)	50 (83)
Qtr6	-328 (63)	45 (191)	34 (113)	37 (82)	17 (90)
Average of 1 to 6	-337 (47)	62 (127)	77 (80)	47 (60)	39 (64)
As a % of impact	77%	142%	176%	107%	88%

^aBootstrap standard errors are shown in parentheses. They are based on 50 replications with 100% resampling.

^bThe estimates for each column are defined as follows:

(1) $\hat{B} = \bar{E}(Y_{0,t}|D=1) - \bar{E}(Y_{0,t}|D=0)$, where $\bar{E}$ denotes the sample mean.

(2) $\hat{B} = \bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t}|D=1) - \bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t}|D=0)$ where $\bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t}|D=1)$ is the sample mean of $(D=1)$ outcomes and $\bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t}|D=0, P)$ the sample mean of nearest neighbor matched ($D=0$) outcomes. The nearest neighbor match for each observation in $(D=1)$ is the observation in $(D=0)$ that is closest in terms of P . Matching is done with replacement. (See Section 3.1 in the text.)

(3) $\hat{B}_{LN} = \bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t}|P \in S_{P,D=1}) - \bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t}|P \in S_{P,D=0})$. Same estimator as (2) except that matches are only constructed within the region of overlapping support S_P , which is precisely defined in Appendix A.

(4) Estimates are constructed using local linear regression on P , as described in the text. There are no variables in the outcome equation. (See Section 5.0 in the text.)

(5) $\hat{B}_{LN,adj} = \bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t} - R\hat{\beta}|P \in S_{P,D=1}) - \bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t} - R\hat{\beta}|P \in S_{P,D=0})$. This is the same estimator as in (4) except matching is performed on the residuals $Y_{0,t} - R\hat{\beta}$ instead of on outcomes $Y_{0,t}$. (See Section 5.0 in the text.) The following regressors R are included in the outcome equation: dummy variables for training center, race, schooling, age, previous training, work experience in months, local unemployment rate, marital status, presence of a child age less than six, and quarter and year effects.

TABLE XVIIIIB
COMPARISON OF ESTIMATED MEAN BIAS
UNDER ALTERNATIVE ESTIMATORS OF MEAN PROGRAM IMPACTS^a
Quarterly Earnings Expressed in Monthly Dollars
Adult Males, 508 Experimental Controls and 388 Elig. Nonparticipants (ENPs)

Quarter	Difference-in-Differences w/o Common Support (1) ^b	Conditional on P Difference-in-Differences w/ Common Support (2)	Regression-Adjusted Conditional on P Difference-in-Differences w/ Common Support (3)
Qtr1	172 (42)	97 (62)	104 (63)
Qtr2	142 (47)	77 (89)	77 (92)
Qtr3	41 (56)	90 (114)	74 (114)
Qtr4	43 (61)	112 (90)	98 (91)
Qtr5	-54 (63)	19 (95)	-5 (99)
Qtr6	-111 (64)	4 (105)	-35 (111)
Average of 1 to 6	39 (47)	67 (71)	52 (74)
As a % of impact	89%	153%	120%

^aBootstrap standard errors are shown in parentheses. They are based on 50 replications with 100% sampling.

^bThe estimates for each column are defined as follows:

(1) $\hat{B}_D = \bar{E}(Y_{0,t}|D=1) - \bar{E}(Y_{0,t}|D=0) - [\bar{E}(Y_{0,t}|D=0) - \bar{E}(Y_{0,t-1}|D=0)]$, where $\bar{E}$ denotes the sample mean.

(2) $\hat{B}_{D,S_P} = \bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t}|P \in S_{P,D=1}) - \bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t}|P \in S_{P,D=0}) - [\bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t-1}|P \in S_{P,D=1}) - \bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t-1}|P \in S_{P,D=0})]$, where $\bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t}|P \in S_{P,D=0})$ is the sample mean of the $D=1$ outcomes, and $\bar{E}_{P|P \in S_{P,D=1}}(Y_{0,t-1}|P \in S_{P,D=0})$ is the sample mean of the $D=0$ matched outcomes. Matches are constructed by local linear regression on P as described in the text (see Section 5.0 in the text). The model does not include regressors in the outcome model.

(3) Same as (2) except the following regressors are included in the outcome equation: training site, age, education, marital status, children less than 6 indicator, indicator for currently enrolled in training, labor market experience, local unemployment rate, season and year.

Similar patterns appear in Table XVIIIIB for the difference-in-differences estimator. Simple differencing symmetrically before and after the date of random assignment or eligibility determination eliminates person-specific components of bias. Compare column (1) of that table with column (1) of Table XVIII. Imposing common support and common density in column (2) generally reduces the quarter-by-quarter bias. However, as we found for the nearest neighbor estimator, the overall average bias is slightly higher. Using regressors to adjust for the bias reduces it slightly as shown in column (3). Note in comparing Tables XVIII and XVIIIIB that the overall bias from our conditional difference-in-differences estimator and from the cross-sectional matching estimator are of the same order of magnitude. Column (3) of Table XVIIIIC reveals that even though the inverse Mills' ratio as typically applied is badly biased (see the estimates in the first column), weighting by a common density ($f(P|D=1)$) greatly improves the performance of the estimator.⁶⁰ Imposing

⁶⁰ For column (3), the ENP observations (for which $D=0$) in the regression are weighted by the ratio $\hat{f}(P|D=1)/\hat{f}(P|D=0)$, where the densities are estimated by standard kernel methods. Imposing the common support condition ensures that the weights are nonzero. The control observations are self-weighting by the $\hat{f}(P|D=1)$ distribution.

TABLE XVIII

COMPARISON OF ESTIMATED MEAN BIAS
UNDER ALTERNATIVE ESTIMATORS OF MEAN PROGRAM IMPACTS^a
Quarterly Earnings Expressed in Monthly Dollars
Adult Males, 508 Experimental Controls and 388 Elig. Nonparticipants (ENPs)

Quarter	Inverse Mills' Ratio w/o Common Support w/o Density Weighting (1) ^b	Inverse Mills' Ratio w/Common Support w/o Density Weighting (2)	Inverse Mills' Ratio w/Common Support w/Density Weighting (3)
Qtr1	-611 (86)	-619 (161)	-147 (176)
Qtr2	-515 (95)	-403 (194)	3 (220)
Qtr3	-498 (96)	-365 (190)	30 (215)
Qtr4	-494 (97)	-421 (191)	-80 (215)
Qtr5	-511 (98)	-441 (190)	-69 (215)
Qtr6	-499 (102)	-323 (196)	48 (222)
Average of 1 to 6	-521 (86)	-553 (161)	-36 (37)
As a % of impact	1198%	985%	83%

^aBootstrap standard errors are shown in parentheses. They are based on 50 replications with 100% sampling.
^bThe estimates for each column are defined as follows:
(1) $\hat{B}_0 = E(\lambda_0(P)|D=1) - E(\lambda_0(P)|D=0)$, where $\hat{E}$ denotes the sample mean, λ_1 is an estimator of $E(U_0|X, D=1)$ obtained under the Mills' ratio assumption and λ_0 is an estimator of $E(U_0|X, D=0)$.
(2) $\hat{B}_{0,CS} = E(\lambda_1(P)|D=1) - E(\lambda_1(P)|D=0)$, where $\hat{E}$ denotes the sample mean, λ_1 is an estimator of $E(U_0|X, D=1)$ and λ_0 is an estimator of $E(U_0|X, D=0)$ obtained under the Mills' ratio assumption. (Same as (1) except mean is only taken over observations in the overlapping support, $S_{0,1}$.)
(3) $\hat{B}_{0,DW} = E(\lambda_1(P)|D=1) - E(\lambda_1(P)|D=0)$, where $\hat{E}$ denotes the sample mean, λ_1 is an estimator of $E(U_0|X, D=1)$ and λ_0 is an estimator of $E(U_0|X, D=0)$ obtained under the Mills' ratio assumption. (Same as (1) except with density weighting as described in the text.)

common support alone without reweighting does not lead to substantial improvement, as shown in column (2).

It is instructive to contrast the biases defined over a common support and with common weighting with the biases defined in the conventional way (e.g., as in Ashenfelter (1978) or LaLonde (1986)). One conventional measure of bias is the OLS estimate of π in the model

$$Y = g(X) + D\pi + U,$$

applied to controls and comparison group members, where $g(X)$ depends on the specification used. The normal selection bias method introduces the inverse Mills' ratio terms into $g(X)$ in conducting a cross-section analysis. The difference-in-differences method uses Y or regression-adjusted Y differenced symmetrically around the date of random assignment or eligibility determination. Estimates of π reveal the bias in the conventional common coefficient model ($U_0 = U_1$), where the program impact is assumed not to depend on X . This estimate of bias combines the three sources of bias distinguished in this paper plus any bias arising from correlation between U_0 and X .⁶¹ In contrast, estimates of the bias that condition on a common support and impose a common weighting of participant and comparison group data produce an estimate of selection bias as rigorously defined in this paper.

⁶¹ Heckman and Todd (1994) decompose the bias π for the model with $g(X) = X\beta$ and present the contribution for the case where U_0 is correlated with X .

(a) Table XVIII

TABLE XIX

COMPARISON OF ESTIMATED MEAN BIAS
UNDER ALTERNATIVE ESTIMATORS OF MEAN PROGRAM IMPACTS^a
Quarterly Earnings Expressed in Monthly Dollars
Adult Males, 508 Experimental Controls and 388 Elig. Nonparticipants (ENPs)

Quarter	Difference in Means (1) ^b	Nearest Neighbor w/o Common Support (2)	Method of Barnow, Cain and Goldberger w/o Common Support w/o Density Weighting (3)	Difference- in-Differences w/o Common Support (4)	Inverse Mills' Ratio w/o Common Support w/o Density Weighting (5)
Qtr1	-418 (38)	221 (56)	-15 (47)	173 (42)	-611 (86)
Qtr2	-349 (47)	-166 (151)	53 (55)	142 (47)	-515 (95)
Qtr3	-337 (55)	-58 (206)	62 (58)	40 (56)	-498 (96)
Qtr4	-286 (57)	161 (178)	107 (60)	43 (61)	-495 (97)
Qtr5	-305 (57)	167 (196)	94 (62)	-54 (63)	-511 (98)
Qtr6	-328 (63)	45 (191)	54 (62)	-111 (64)	-499 (102)
Average of 1 to 6	-337 (47)	62 (127)	62 (58)	39 (47)	-521 (86)
As a % of impact	775%	143%	143%	90%	1198%

^aBootstrap standard errors are shown in parentheses. They are based on 50 replications with 100% sampling.
^bThe estimates for each column are defined as follows:
(1) $\hat{B}_0 = E(Y_0|D=1) - E(Y_0|D=0)$, where $\hat{E}$ denotes the sample mean.
(2) $\hat{B}_0 = E(Y_0|D=1) - E(Y_0|D=0)$, where $\hat{E}_{NP,D=1}(Y_0|D=1)$ is estimated by the sample mean of $(D=1)$ outcomes and $\hat{E}_{NP,D=0}(Y_0|D=0)$ by the sample mean of $(D=0)$ nearest neighbor matches. (See Section 3.1 in the text.)
(3) Same as described in footnote 4 in Table XII except without imposing a common support restriction.
(4) $\hat{B}_0 = E(Y_{0,1}|D=1) - E(Y_{0,-1}|D=1) - (E(Y_{0,1}|D=0) - E(Y_{0,-1}|D=0))$, where $\hat{E}$ denotes the sample mean.
(5) $E(\lambda_1(P)|D=1) - E(\lambda_1(P)|D=0)$, where $\hat{E}$ denotes the sample mean, λ_1 is the estimator for $E(U_0|X, D=1)$ under the Mills' ratio assumption, and λ_0 is the estimator of $E(U_0|X, D=0)$.

The estimates of π for the different methods are presented in Table XIX. Except for the inverse Mills' ratio, the overall biases (π) from the other commonly-used estimators are of the same order of magnitude. All except the inverse Mills' ratio estimator produce biases that are smaller than the raw mean $\hat{B}$. At the same time, all are large relative to the program impact and exhibit substantial variability across quarters. The different sources of bias tend to cancel each other out. This is especially true of the Barnow, Cain, and Goldberger (1980) estimator. (Compare Column (3) of Table XIX with the last column of Table XII).

By decomposing the bias π into its components, we determine whether a small estimated π is due to a fortuitous combination of offsetting biases or whether each component of the bias is small. Sources of bias such as the failure of common support and discrepancies in the weights across participants and comparison group members depend on the sampling plan used to collect the data for the comparison group and so are likely to vary across evaluations. The factors generating self-selection are more likely to be similar across evaluations. The focus in this paper is on the estimation of the stable components of the conventional measure of bias. Knowledge of these components facilitates generalization of the evidence from any one study to other environments, and is more

(b) Table XIX

6 Interpretive synthesis

Matching reduces some bias but fails as a general identifying strategy. Selection bias is not the same as raw mean differences. Data quality, geography, questionnaire comparability, support, and density weighting are all part of identification. Conditional DID is promising when selection bias is stable over time after conditioning on P .

7 Short summary

- The paper uses experimental data to measure selection bias directly.
- The key bias object is $B(X) = E(Y_0 | X, D = 1) - E(Y_0 | X, D = 0)$.
- Raw bias decomposes into support mismatch, weighting mismatch, and true selection bias.
- Matching on P reduces but does not eliminate bias.
- The normal inverse-Mills model is not supported by the estimated bias function.
- Index sufficiency and conditional DID are more promising, with important qualifications.
- Same geography, same questionnaire, and rich recent labor-force histories are essential for good comparison data.

8 Conclusion

8.1 Core conclusion

Randomized experimental data can be used to diagnose nonexperimental estimators. The paper rejects simple matching assumptions, supports more structured semiparametric approaches, and emphasizes data quality.

8.2 Economic intuition

Selection bias is generated by participation decisions that depend on recent labor-market states and local conditions. Some of this can be matched; some must be modeled; some can be removed by differencing if stable over time.

8.3 Takeaway

Credible program evaluation requires well-designed comparison data, common support, correct weighting, rich covariates, and direct tests of identifying assumptions.